

Is a structural-MRI foundation model worth its weight?

A four-rung classical morphometry ladder for the Dallas Lifespan Brain Study sMRI-FM benchmark

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Draft — April 24, 2026

Abstract

The release of the MedARC `smri-fm` dataset, the publication of BrainIAC [AIM-KannLab contributors, 2026], and the MICCAI 2025 FOMO25 and SSL3D challenges won by Gordaliza et al. [2026] have renewed interest in structural-MRI foundation models (sMRI-FMs) as a general-purpose backbone for neuroimaging. Whether FMs provide a meaningful advance over classical morphometric features for core tasks — above all, brain-age prediction — has not been settled. In fact Schulz et al. [2025] show that for *disease detection*, simpler models beat complex ones despite losing on age MAE: ridge regression with 1 400 parameters produces larger patient-vs-control effect sizes than a 46M-parameter CNN on the same data. We therefore construct a **four-rung classical morphometry ladder** on the Dallas Lifespan Brain Study (DLBS, Park et al., 2025) and benchmark three sMRI foundation-model checkpoints against it under identical bias-correction regimes. This manuscript is generated with Pweave so every number and figure below updates as the experiment parquets land.

Current snapshot: 23 subjects, 60 sessions processed by FastSurfer VINN; 23 subjects, 60 sessions by T1Prep (CAT-Surface).

1 Introduction

Brain-predicted age, introduced by Franke et al. [2010] and popularised by Cole and colleagues [Cole et al., 2017], has become the single most-reported biomarker derived from structural MRI. A decade of architectural progress — from relevance-vector regression on VBM features, through Gaussian-process regression [Cole et al., 2017, Cole] to 3D CNNs [Peng et al., 2021] — has pushed UK Biobank mean absolute error (MAE) from ~ 5 years to ~ 2 years [Peng et al., 2021]. At the same time, multiple authors have shown that classical ridge- or kernel-based regressors on FreeSurfer or CAT12 morphometric features match deep networks within noise [More et al., 2023, Lombardi et al., 2021, Bashyam et al., 2020]. Gijssen et al. [2024] established in a 46 381-subject UK Biobank study that ResNet-50 and SwinT reach an indistinguishable 2.66 years MAE, with SwinT scaling better only above $\sim 25k$ training subjects. The apparent ceiling is set by label noise and site effects, not by model capacity.

Schulz et al. [2025] take this further and show that for disease detection, simpler models actually *beat* complex ones on patient-vs-control effect size even when they lose on age MAE. This finding motivates our evaluation plan: disease-effect-size is reported alongside age MAE for every arm.

2 Methods (pre-registration)

2.1 Data

We use the Dallas Lifespan Brain Study [Park et al., 2025], OpenNeuro `ds004856`, with 464 / 338 / 224 subjects across waves 1/2/3 (~ 1000 T1 sessions, Philips 3T Achieva, ~ 4 –5 year intervals, ages 20 to 90 at wave 1). External generalisation is evaluated on `ds003592` [Spreng et al., 2022]. Both were downloaded from OpenNeuro S3 on 2026-04-22.

2.2 Four classical morphometry rungs

Every pipeline runs inside a pinned Docker image from the `ghcr.io/m9h/neurocontainers-arm` registry for reproducibility on NVIDIA Grace-Blackwell arm64 hosts.

Rung 1 — SynthSeg volumes (MedARC baseline). The MedARC `preprocessing/pipeline.py` [MedARC-AI contributors, 2026]: `SynthStrip`, ANTs rigid to MNI152NLin2009cAsym, `mri_synthseg --parc --robust`. Aggregated to 71 ICV-normalised features per Bethlehem et al. [2022]. Implemented upstream by Ashtari [2026].

Rung 2 — SynthSeg + N4. Tests whether an upstream bias step helps a bias-robust segmentation network.

Rung 3 — FastSurfer VINN. `aparc.DKTatlas+aseg.deep` VINN + CerebNet + HypVINN. Isolates seg-network quality from bias handling. Runs in `ghcr.io/m9h/fastsurfer-arm:6b6b985-arm.1`.

Rung 4 — T1Prep thickness + Jacobian. T1Prep v0.3.3 [Gaser et al., 2026] (DeepMriPrep + AMAP + CAT-Surface). Adds cortical thickness per Desikan–Killiany ROI and nonlinear-warp log-Jacobian — the two feature axes no volumetric seg captures. Runs in `ghcr.io/m9h/t1prep-arm:v0`.

2.3 Foundation-model arms

Arm	Design	Reference
BrainIAC	SimCLR ViT, 32k pretraining scans	[AIM-KannLab contributors, 2026]
FOMO25 mmunetvae	3D U-Net MAE, 11 187 FOMO-60K scans	[Gordaliza et al., 2026]
AnatCL	anatomy-supervised contrastive, OpenBHB 3 984	[Barbano et al., 2024]
MedARC smri-fm	TBD; reported when release lands	[MedARC-AI contributors, 2026]

2.4 Bias correction

For every arm we report MAE under four regimes: raw, Cole-Smith [Smith et al., 2019], Beheshti [Liang et al., 2019], and Zhang age-level [Zhang et al., 2023].

2.5 Disease-effect-size reporting per Schulz et al. [2025]

For each arm we report patient-vs-control effect size (Cohen’s d) on an AD-spectrum labelling derived from DLBS participant metadata.

2.6 Longitudinal stability per Vidal-Piñeiro et al. [2021]

For each returning subject we compute $\Delta\text{BAG}_{i,t}$ between waves and test (a) within-subject stability (ICC), and (b) correlation with change in morphometric summaries. DLBS’s three-wave structure is the first public venue where this test can be run on an sMRI-FM.

3 Current results

3.1 Dataset coverage

Tool	Subjects	Sessions	Feature rows
FastSurfer VINN	23	60	9900
T1Prep	23	60	7934

3.2 FastSurfer–T1Prep concordance on sub-1003

Figure 1: DK ROI volume (FastSurfer VINN) vs cortical thickness (T1Prep CAT-Surface) for DLBS sub-1003 across three waves.

3.3 Ridge brain-age baseline (current parquet)

Arm & correction	n scans	MAE (yr)	RMSE	R^2	Pearson r
FastSurfer VINN (rung 3) / raw	60	6.61	8.35	0.82	0.91
FastSurfer VINN (rung 3) / cole	60	7.60	9.19	0.79	0.91
FastSurfer VINN (rung 3) / beheshti	60	6.16	7.45	0.86	0.94
FastSurfer VINN (rung 3) / zhang	60	5.25	6.83	0.88	0.95
T1Prep thickness (rung 4) / raw	60	9.23	11.63	0.66	0.82
T1Prep thickness (rung 4) / cole	60	10.73	13.68	0.52	0.82
T1Prep thickness (rung 4) / beheshti	60	8.27	10.54	0.72	0.88
T1Prep thickness (rung 4) / zhang	60	7.98	10.27	0.73	0.88
BrainIAC 768-d (FM) / raw	60	14.44	17.01	0.27	0.55
BrainIAC 768-d (FM) / cole	60	24.79	29.97	-1.28	0.55
BrainIAC 768-d (FM) / beheshti	60	10.27	12.41	0.61	0.85
BrainIAC 768-d (FM) / zhang	60	9.61	11.95	0.64	0.85
FS+T1Prep concat / raw	60	6.57	8.04	0.84	0.91
FS+T1Prep concat / cole	60	7.46	8.78	0.80	0.91
FS+T1Prep concat / beheshti	60	6.16	7.26	0.87	0.94
FS+T1Prep concat / zhang	60	6.17	7.24	0.87	0.94
FS+T1Prep+BrainIAC concat / raw	60	7.40	8.83	0.80	0.90
FS+T1Prep+BrainIAC concat / cole	60	7.96	9.84	0.75	0.90
FS+T1Prep+BrainIAC concat / beheshti	60	6.53	8.08	0.83	0.93
FS+T1Prep+BrainIAC concat / zhang	60	6.60	8.10	0.83	0.92

4 Infrastructure contributions

All pipelines run on an NVIDIA DGX Spark (GB10 Grace-Blackwell, arm64). Upstream neuroimaging containers are almost entirely amd64 and fall back to QEMU emulation on Grace, which is unusably slow for TF/PyTorch inference. To close this gap we contribute:

- `ghcr.io/m9h/t1prep-arm:v0.3.3-arm.1` — arm64 T1Prep with three Blackwell-specific patches pending upstream PR (deterministic-algorithm warn_only fallback, weight-cache write permissions, case-sensitive Neuromorphometrics atlas filenames).
- `ghcr.io/m9h/fastsurfer-arm:6b6b985-arm.1` — arm64 FastSurfer seg_only, NGC PyTorch 24.12 base. Published in response to Deep-MI/FastSurfer issue #716.
- `ghcr.io/m9h/medarc-arm` (pending build) — MedARC’s `smri-fm/preprocessing/pipeline.py` on arm64 via locally built FreeSurfer 8.2.0 .debs.

Recipes + auto-build workflow at <https://github.com/m9h/neurocontainers-arm>.

5 Pre-registered hypotheses

1. *Cross-sectional ceiling*. FM + ridge and rung-4 (T1Prep thickness + Jacobian) + ridge match within 0.3 years MAE on wave-1 held-out subjects.
2. *Bias persistence*. Regression-dilution slope α indistinguishable across preprocessing and FM arms.
3. *Longitudinal disconnect*. Within-subject Δ BAG correlates weakly ($|r| < 0.2$) with within-subject morphometric change, per Vidal-Piñero et al. [2021], for every arm.
4. *Disease-effect-size inversion*. Per Schulz et al. [2025], rungs 1–4 have higher patient-vs-control effect size than any FM + ridge despite losing on age MAE.

6 Discussion (placeholder)

To be written after ridge numbers land. The contribution is not a new model but an apples-to-apples evaluation that pins down where the sMRI-FM proposition is and is not load-bearing.

References

- AIM-KannLab contributors. A generalizable foundation model for analysis of human brain MRI. *Nature Neuroscience*, 2026. doi: 10.1038/s41593-026-02202-6. Preprint: <https://www.medrxiv.org/content/10.1101/2024.12.02.24317992v1>.
- Nima Ashtari. SynthSeg ridge baseline for brain-age prediction on DLBS. https://github.com/MedARC-AI/smri-fm/tree/main/experiments/brainage_synthseg_ridge, 2026. Merged as PR #9.
- Carlo Alberto Barbano, Matteo Brunello, Benoit Dufumier, and Marco Grangetto. Anatomical foundation models for brain MRIs. *arXiv*, 2408.07079, 2024.

- Vishnu M. Bashyam, Guray Erus, Jimit Doshi, Mohamad Habes, Ilya Nasrallah, Monica Truelove-Hill, Dhivya Srinivasan, Lisa Mamourian, Raymond Pomponio, Yong Fan, et al. MRI signatures of brain age and disease over the lifespan based on a deep brain network and 14 468 individuals worldwide. *Brain*, 143(7):2312–2324, 2020. doi: 10.1093/brain/awaa160.
- R. A. I. Bethlehem, J. Seidlitz, S. R. White, J. W. Vogel, K. M. Anderson, C. Adamson, S. Adler, G. S. Alexopoulos, E. Anagnostou, A. Areces-Gonzalez, et al. Brain charts for the human lifespan. *Nature*, 604:525–533, 2022. doi: 10.1038/s41586-022-04554-y.
- James H. Cole. brainageR: Software for generating a brain-predicted age value using Gaussian processes regression. <https://github.com/james-cole/brainageR>. Accessed 2026-04-21.
- James H. Cole, Rudra P.K. Poudel, Dimosthenis Tsagkrasoulis, Matthan W.A. Caan, Claire Steves, Tim D. Spector, and Giovanni Montana. Predicting brain age with deep learning from raw imaging data results in a reliable and heritable biomarker. *NeuroImage*, 163:115–124, 2017. doi: 10.1016/j.neuroimage.2017.07.059.
- Katja Franke, Gabriel Ziegler, Stefan Klöppel, Christian Gaser, and Alzheimer’s Disease Neuroimaging Initiative. Estimating the age of healthy subjects from T1-weighted MRI scans using kernel methods: Exploring the influence of various parameters. *NeuroImage*, 50(3):883–892, 2010. doi: 10.1016/j.neuroimage.2010.01.005.
- Christian Gaser, Robert Dahnke, et al. T1Prep: Python-based preprocessing for T1-weighted MRI. 2026. Accessed 2026-04-21.
- Sam Gijzen et al. Do transformers and CNNs learn different concepts of brain age? *Human Brain Mapping*, 45:e70243, 2024. doi: 10.1002/hbm.70243.
- Pedro M. Gordaliza, Jaume Banus, Benoît Gérin, Maxence Wynyen, Nataliia Molchanova, Jonas Richiardi, and Meritxell Bach Cuadra. From 100,000+ images to winning the first brain MRI foundation model challenges: Sharing lessons and models. *arXiv*, 2601.13166, 2026. Won MICCAI 2025 SSL3D and FOMO25 challenges.
- Hualou Liang, Fengqing Zhang, and Xin Niu. Investigating systematic bias in brain age estimation with application to post-traumatic stress disorders. *Human Brain Mapping*, 40(11):3143–3152, 2019. doi: 10.1002/hbm.24588.
- Angela Lombardi, Nicola Amoroso, Domenico Diacono, Alfonso Monaco, Sabina Tangaro, and Roberto Bellotti. Extensive evaluation of morphological statistical harmonization for brain age prediction. *Brain Sciences*, 11(6):788, 2021. doi: 10.3390/brainsci11060788.
- MedARC-AI contributors. MedARC structural MRI foundation model project. <https://github.com/MedARC-AI/smri-fm>, <https://huggingface.co/datasets/medarc/smri-fm>, 2026. Open community project; kickoff 2026-04-09.
- Shammi More, Georgios Antonopoulos, Felix Hoffstaedter, Julian Caspers, Simon B. Eickhoff, and Kaustubh R. Patil. Brain-age prediction: A systematic comparison of machine learning workflows. *NeuroImage*, 270:119947, 2023. doi: 10.1016/j.neuroimage.2023.119947.
- Denise C. Park, Joseph Hennessee, Evan T. Smith, Micaela Chan, Carson Katen, Julia Bacci, Sarah Frank, Sarah Monier, Alexandra Collyer, Carol Tamminga, William Moore, Neil Rofsky, Karen Rodrigue, Kristen Kennedy, and Gagan Wig. The Dallas Lifespan Brain Study: A comprehensive

- adult lifespan data set of brain and cognitive aging. *Scientific Data*, 12:846, 2025. doi: 10.1038/s41597-025-04847-7.
- Han Peng, Weikang Gong, Christian F. Beckmann, Andrea Vedaldi, and Stephen M. Smith. Accurate brain age prediction with lightweight deep neural networks. *Medical Image Analysis*, 68: 101871, 2021. doi: 10.1016/j.media.2020.101871.
- Marc-Andre Schulz, Simon Siegel, and Kerstin Ritter. Brain-age models with lower age prediction accuracy have higher sensitivity for disease detection. *PLOS Biology*, 23(x):e3003451, 2025. doi: 10.1371/journal.pbio.3003451.
- Stephen M. Smith, Diego Vidaurre, Fidel Alfaro-Almagro, Thomas E. Nichols, and Karla L. Miller. Estimation of brain age delta from brain imaging. *NeuroImage*, 200:528–539, 2019. doi: 10.1016/j.neuroimage.2019.06.017.
- R. Nathan Spreng, Roni Setton, Udi Alter, Benjamin N. Cassidy, Bri Darboh, Elizabeth DuPre, Karin Kantarovich, Amber W. Lockrow, Laetitia Mwilambwe-Tshilobo, Wen-Ming Luh, Prantik Kundu, and Gary R. Turner. Neurocognitive aging data release with behavioral, structural and multi-echo functional MRI measures. *Scientific Data*, 9:119, 2022. doi: 10.1038/s41597-022-01231-7. OpenNeuro ds003592.
- Dídac Vidal-Piñeiro, Yunpeng Wang, Stine K. Krogsrud, Inge K. Amlien, William F.C. Baaré, David Bartrés-Faz, Lars Bertram, Andreas M. Brandmaier, Christian A. Drevon, Sandra Düzel, et al. Individual variations in ‘brain age’ relate to early-life factors more than to longitudinal brain change. *eLife*, 10:e69995, 2021. doi: 10.7554/eLife.69995.
- Biao Zhang, Shuqi Zhang, Jianfeng Feng, and Shu Zhang. Age-level bias correction in brain age prediction. *NeuroImage: Clinical*, 37:103319, 2023. doi: 10.1016/j.nicl.2023.103319.